

Ernest Essuah Mensah

Senior Product Engineer | Production AI & Agent Systems

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SUMMARY

Senior product engineer who builds and ships reliable systems end to end, with particular depth in production LLM applications and the agent-ready infrastructure that makes AI-assisted development safe, observable, and verifiable.

EXPERIENCE

Sole Product Engineer — Athena

Mar 2026 – Present

Business operating system for retail and service businesses, in production for Wigclub (Ghana)

- Built and operate the full TypeScript system across point-of-sale, online storefront, inventory, payments, and service operations; production now manages 1,800+ SKUs and has processed 1,100+ completed POS transactions.
- Architected the POS local-first: the register reads and writes to on-device storage so checkout never blocks on the network, with a sync layer that reconciles sales to the backend and preserves ordering across concurrent registers.
- Built an agent-ready delivery harness spanning 35 validation surfaces across three apps, combining generated repo maps, deterministic and LLM-based reviews, runtime evidence, and drift sensors; a recorded five-phase merge-grade run completed in under 10 minutes with zero duplicate commands.
- Built the deployment pipeline: one-command deploys to a VPS behind Cloudflare Tunnel, automatic QA deploys on every merge, and instant rollback to any previous release.

Product & AI Engineer — kwamina.fyi (Independent Project)

2026

React 19, Cloudflare Workers, D1, Claude

- Built a site assistant grounded by construction: published pages compile into a versioned, prompt-cached corpus, so it can never claim anything a visitor can't read; vector retrieval deliberately deferred behind a measured token budget.
- Engineered honest streaming: tokens render immediately, but the success signal waits for the database commit; history is server-authoritative so a tampered client can't rewrite what the model sees.
- Privacy-first observability (one sanitization contract across browser, worker, and analytics; allowlisted telemetry) held by ~200 tests, including a Python suite validating the content itself, and a scheduled production canary.

Senior Software Engineer — Eventbrite

Mar 2022 – 2026 · Remote

Promoted SWE I → SWE II (2023) → Senior (2025)

- Technical lead and main implementer of Dashy, Eventbrite's AI-powered Event Dashboard Assistant (Amazon Bedrock). Owned it end-to-end: hardened chat BFF routes (auth, validation, rate limiting, streaming), lifecycle-grounded prompts and guardrails, and CSAT collection.
- Pioneered an agentic AI development workflow that scaffolded requirements, tickets, and code, cutting feature delivery from weeks to days; taught it at internal engineering forums.
- Core engineer on the migration of the legacy Event Dashboard to a modern Next.js stack: routing, permissions, analytics parity, and feature-flagged rollout with documented rollback. Built the error-recovery layer that eliminated 8+ second full-page reloads during backend failures.
- Led fraud prevention across subscription and checkout payment flows with Stripe Radar, cutting fraudulent payments 65% within three months; implemented 3D Secure/SCA verification, unblocking subscriptions for customers across the EU and India.
- Championed nonprofit plan automation beyond the team's roadmap, replacing an error-prone manual back-office process and auto-enrolling ~3,200 verified NPOs into discounted plans at launch.
- Built core Stripe billing services and infrastructure (Terraform-provisioned API Gateway with WAF, Lambda, DynamoDB, webhook processing, refund and proration endpoints), raising payment-service test coverage from 58% to 90%+.
- Shipped a logged-out event-creation flow that removed account signup as a prerequisite to building an event, lifting event creation +26.6%, event publishing +5.4%, and transactions +2.2%.

Apple, Software Engineering Intern (Jun – Nov 2021): migrated Java API test suites to Karate, cutting build and test times by 20%; expanded coverage of OS-services APIs by 40%+.

University of Maryland iSchool, Research Software Engineer (Dec 2019 – May 2021): built the UI of a teachable object recognizer helping blind users identify objects; co-authored papers presented at ACM ASSETS and CHI.

EDUCATION

Bachelor of Science in Computer Science, University of Maryland, College Park

May 2021

SKILLS

Application engineering: TypeScript, Python, Node.js, React, Next.js, Tailwind, Tanstack · **AI systems:** Production LLM applications, grounding and context engineering, prompt engineering and guardrails, streaming, LLM evaluation, AI observability · **Agent engineering:** Agent-ready repositories, multi-agent orchestration, tool and skill design, deterministic and LLM-based review harnesses · **Payments & risk:** Stripe, Radar, 3D Secure/SCA · **Cloud, data & delivery:** AWS, Cloudflare Workers/D1, Convex, PostgreSQL, Valkey/Redis, Terraform,

